Hydrophones and Ocean Sound Propagation

How do we listen to sound travelling through the ocean?

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https://www.oceannetworks.ca/multimedia/sounds/

Learning objectives

By the end of this unit, you should be able to:

- Explain how sound propagates underwater.
- Describe the main controls on sound speed in seawater.
- Explain how a hydrophone records underwater pressure fluctuations.
- Describe refraction and the SOFAR channel.
- Interpret simple underwater sound-propagation paths.

How fast does sound travel underwater?

Sound travels much faster in water than in air.

- Seawater: approximately 1500 m s^{-1}
- Air: approximately 343 m s^{-1}

The exact speed in seawater is not constant. It changes with the physical properties of the water.

Think about it

Why might efficient sound propagation make acoustics particularly useful for monitoring the ocean?

$$Time\ (s) = \frac{Distance\ (m)}{Speed\ (\frac{m}{s})}$$

Quick check

Approximately how long would sound take to travel 15 km through seawater?

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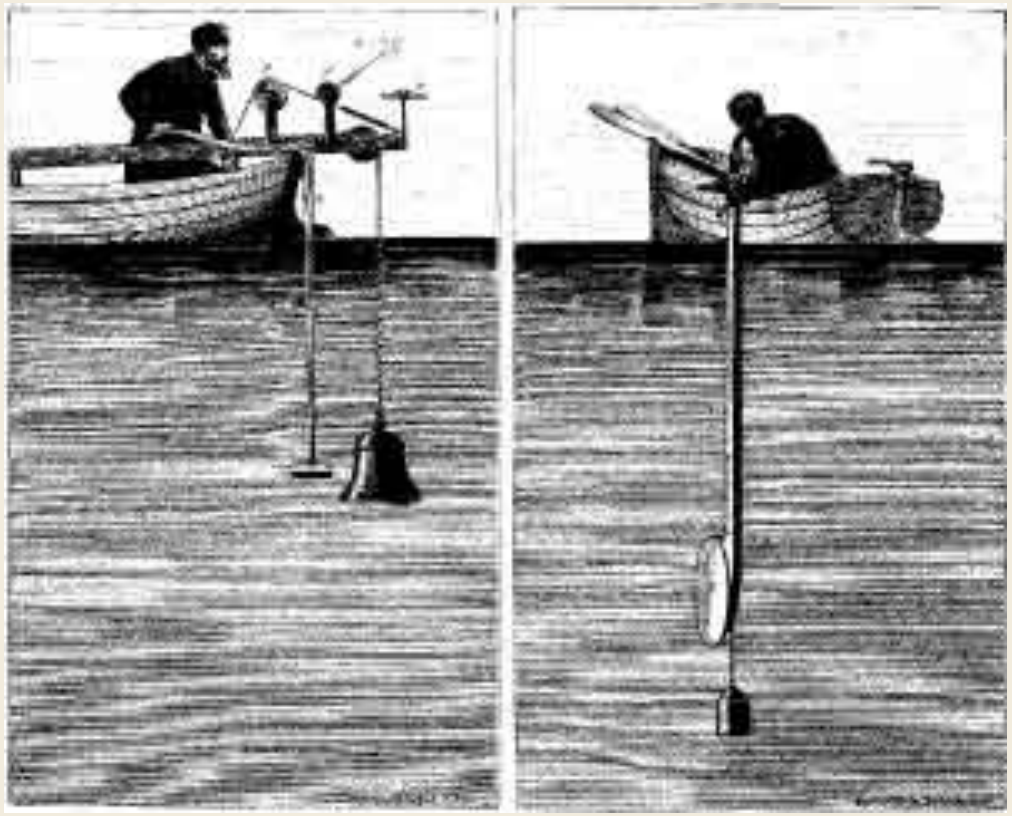
Answer: ~10 s

What controls sound speed in seawater?

Three important factors control the speed of sound:

- **Temperature**
  Warmer water generally has a higher sound speed.
- **Salinity**
  More saline water generally has a higher sound speed.
- **Pressure / depth**
  Increasing pressure with depth generally increases sound speed.

The sound speed at any location therefore depends on the combined effect of these properties.



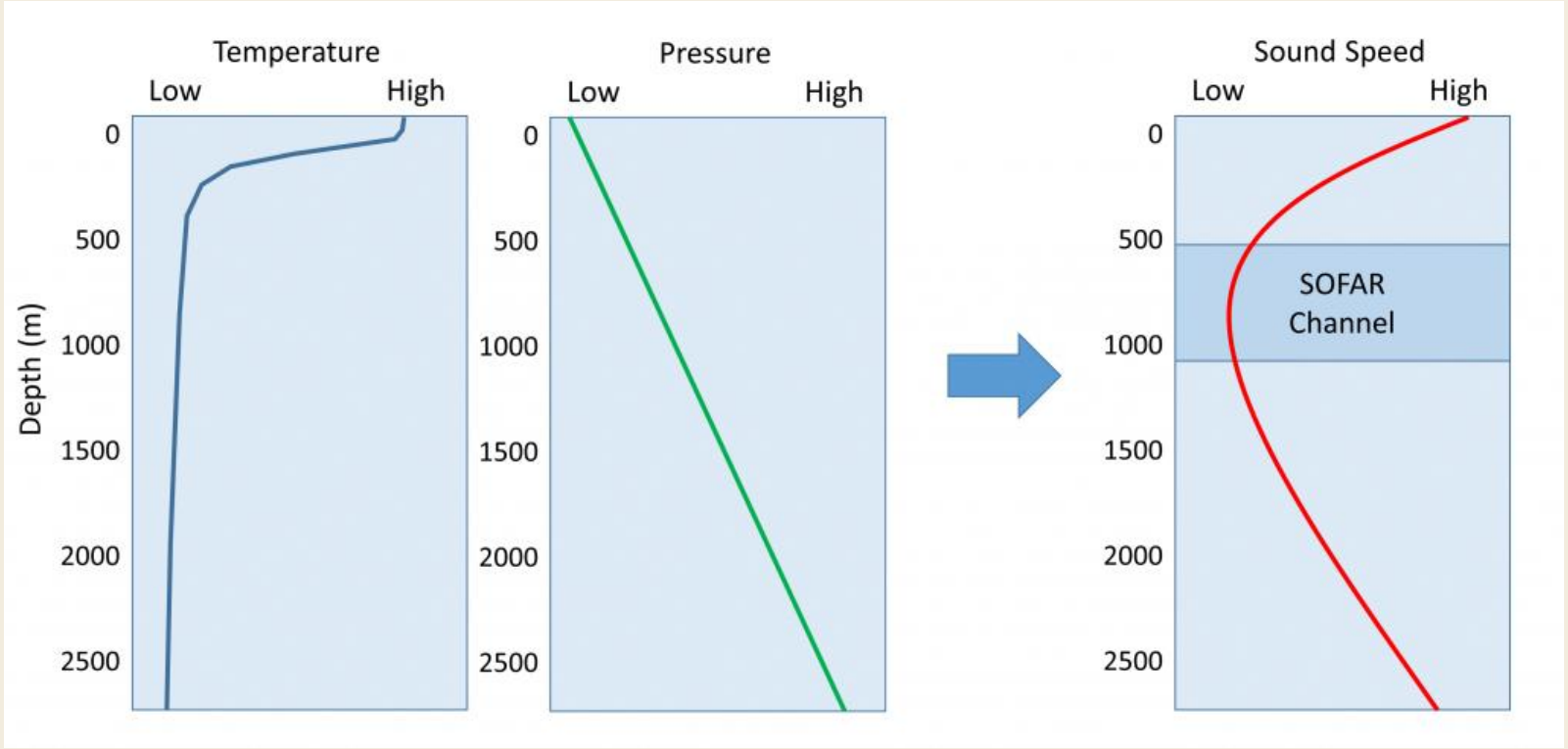
Did you know?

In 1826, Jean-Daniel Colladon and Charles-François Sturm measured the speed of sound in Lake Geneva. They struck an underwater bell while simultaneously producing a visible flash. By measuring the delay between seeing the flash and hearing the bell approximately 16 km away, they estimated the speed of sound in water with remarkable accuracy.
Source: https://dosits.org/decision-makers/tutorials/science/sound-speed

Sound-speed profile through the ocean

Sound speed changes with depth. A typical deep-ocean profile may contain:

- **Near the surface**
  Relatively warm water can produce high sound speed.
- **Thermocline**
  Temperature decreases rapidly with depth, often causing sound speed to decrease.
- **Deep ocean**
  Increasing pressure becomes increasingly important, causing sound speed to increase again.

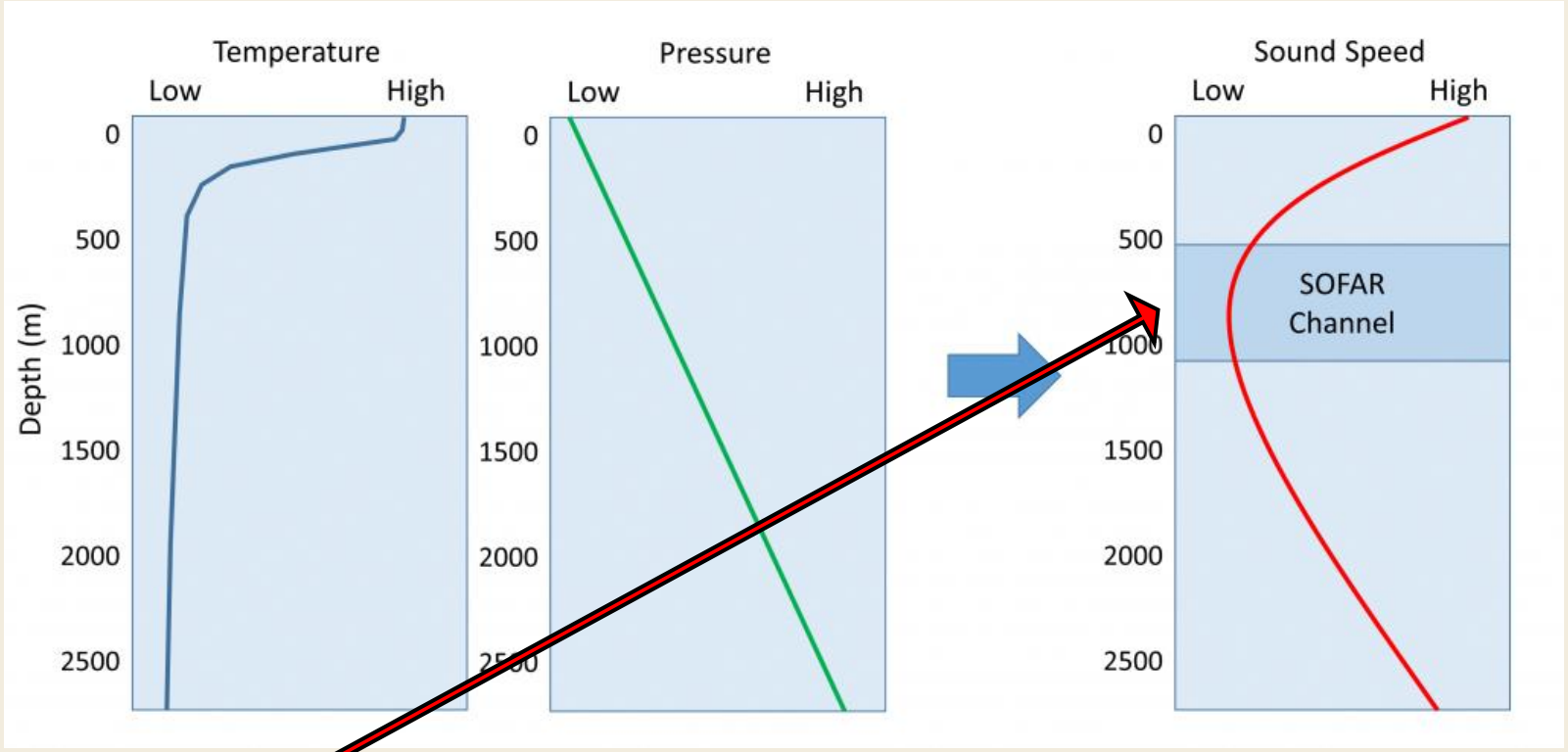


Source: Webb (2023), Introduction to Oceanography, Fig. 6.4.1, CC BY 4.0.

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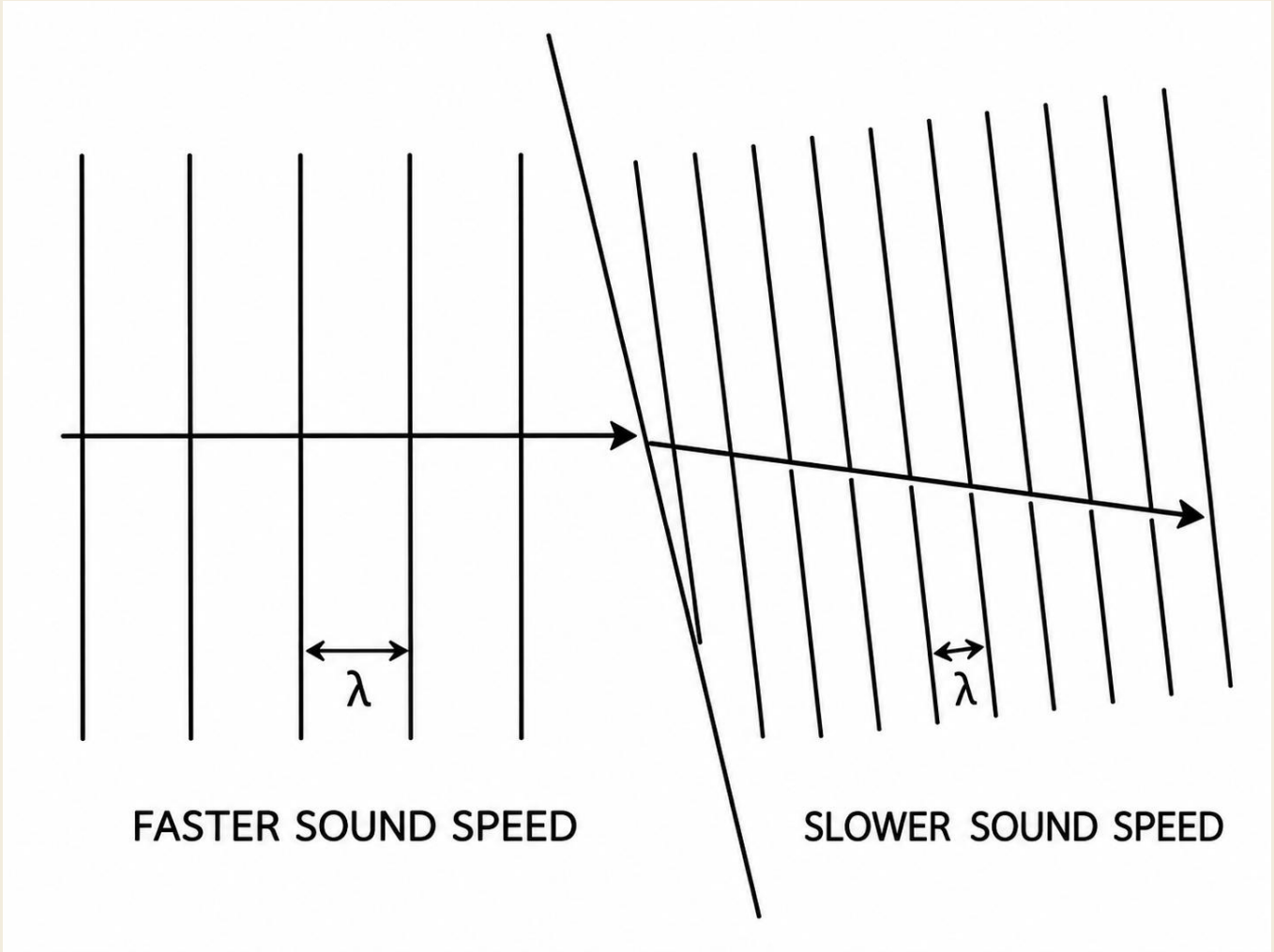
This can produce a sound-speed minimum at intermediate depth.

Sound does not always travel in a straight line

When sound speed changes through the ocean, sound waves are refracted.

Refraction
The change in the direction of propagation of a wave when wave speed varies spatially

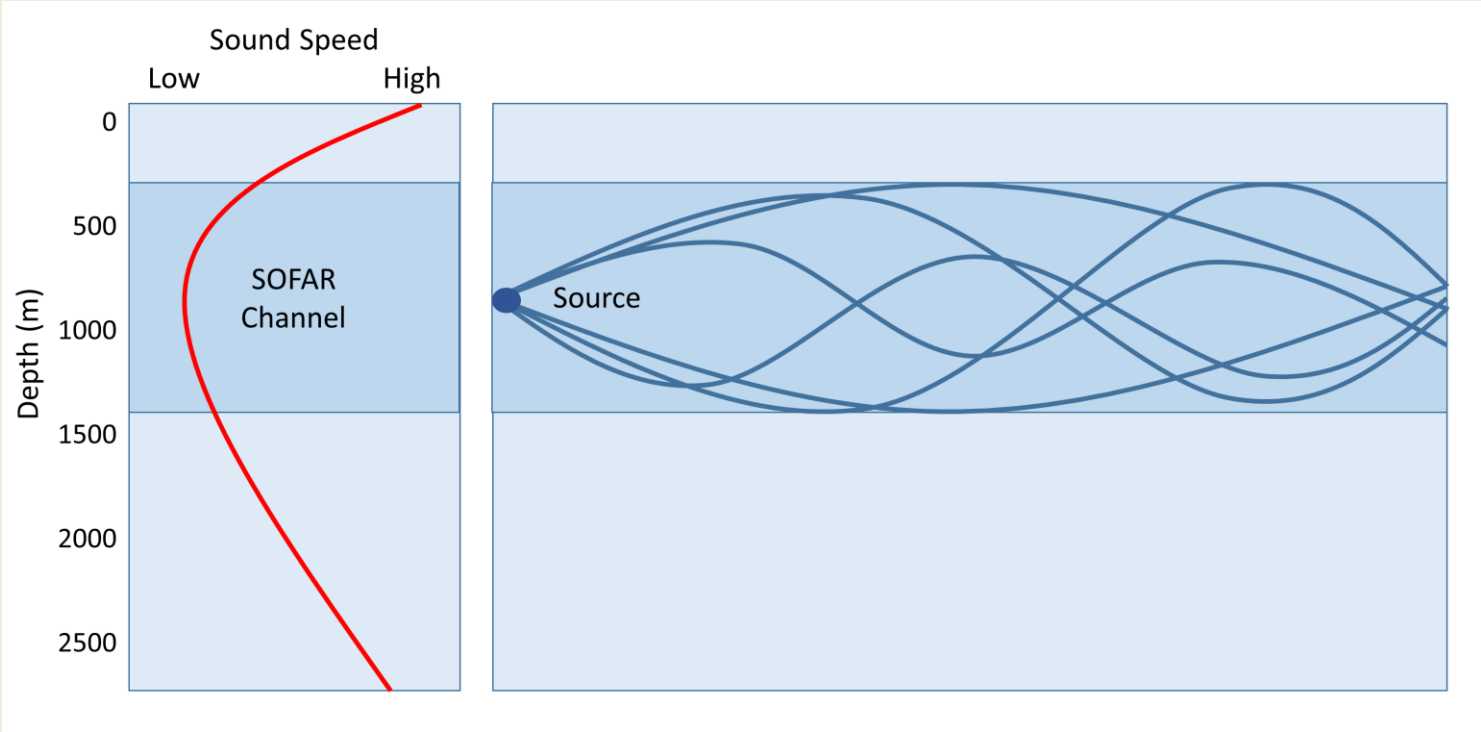
Sound rays tend to bend towards regions of lower sound speed. Therefore, vertical changes in sound speed can cause sound to curve through the ocean rather than travelling in a straight line.



The SOFAR channel
SOFAR = Sound Fixing and Ranging

In many parts of the deep ocean, sound speed reaches a minimum at intermediate depth.
Sound travelling away from this minimum enters regions of higher sound speed and is refracted back towards the minimum.
The result is an acoustic waveguide that can trap and guide sound over very long distances.

The depth of the SOFAR channel varies geographically.



Source: Webb (2023), Introduction to Oceanography, Fig. 6.4.1, CC BY 4.0.

Typical mid-latitude depth: ~1000 m, but the channel depth varies geographically.

Why can SOFAR signals travel so far?

Sound travelling within the SOFAR channel is repeatedly refracted back towards the channel axis.

This can reduce interaction with:

- the sea surface
- the seabed

Reducing these losses allows some low-frequency sounds to propagate over very large distances.

Under suitable ocean conditions, signals can travel hundreds to thousands of kilometres!

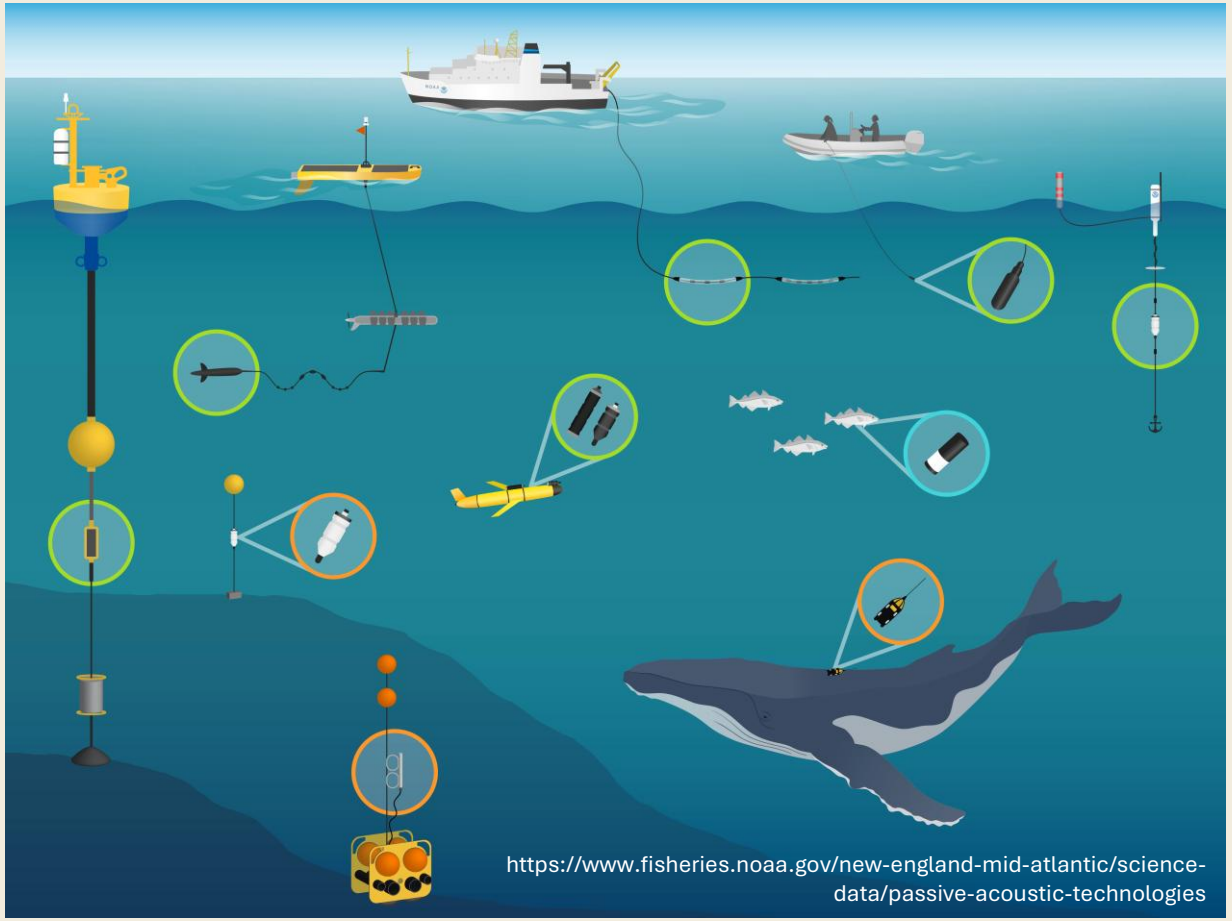


Long-range propagation is especially effective for low-frequency sound.
Low-frequency sound is absorbed less strongly by seawater than high-frequency sound.

Applications of long-range underwater sound

Long-distance acoustic propagation allows scientists to detect and study events far from the receiver. Examples include:

- **Marine mammals**
  Long-range vocalisations and passive acoustic monitoring.
- **Submarine earthquakes**
  Hydroacoustic signals generated by seismic activity.
- **Volcanic activity**
  Detection of submarine eruptions and associated acoustic signals.
- **Nuclear-test monitoring**
  Hydroacoustic stations form part of international monitoring systems.
- **Ocean monitoring**
  Persistent acoustic observations can provide information over large spatial scales.



What is a hydrophone?

A hydrophone is an underwater acoustic sensor. It detects changes in water pressure associated with sound waves.
The hydrophone converts these pressure fluctuations into an electrical signal that can be recorded and analysed.



Underwater sound → pressure fluctuations → hydrophone → electrical signal

From underwater pressure to digital data

A hydrophone recording passes through several stages:

UNDERWATER SOUND
pressure fluctuations
↓
HYDROPHONE
pressure → analogue voltage
↓
RECORDER / ADC
analogue → digital
↓
DIGITAL TIME SERIES
samples stored through time
↓
ANALYSIS
waveform · spectrum · spectrogram

Two important recording properties

Sampling frequency
number of samples recorded each second.

Recording duration
total time over which the signal is recorded.

Where can hydrophones be deployed?

- **Seabed**
  Stable location and suitable for long-term monitoring.

- **Mooring**
  Hydrophone suspended within the water column.

- **Drifting or autonomous platform**
  Moves through the ocean and can sample different locations.

- **Cabled observatory**
  Connected to shore for power and continuous data transmission.



Each approach involves trade-offs in:

- deployment duration
- spatial coverage
- power
- data recovery
- cost

Did you know?

The Comprehensive Nuclear-Test-Ban Treaty Organization (CTBTO) operates a global hydroacoustic monitoring network comprising six underwater hydrophone stations and five land-based seismic T-phase stations.

Its underwater hydrophones are deployed at depths of approximately 600–1,200 m, near the SOFAR channel axis, depending on local ocean conditions.

These stations detect low-frequency acoustic signals generated by distant natural events and human activities, supporting global nuclear-test monitoring.

CTBTO
PREPARATORY COMMISSION



https://www.ctbto.org/our-work/monitoring-technologies/hydroacoustic-monitoring

Take-home messages

- Sound travels at ~1500 m s^{-1} in seawater, but sound speed varies.
- Temperature, salinity and pressure control sound speed.
- Sound-speed gradients cause refraction.
- A sound-speed minimum can form the SOFAR channel.
- Low-frequency sound can travel extremely long distances.
- Hydrophones convert underwater pressure fluctuations into electrical signals, which recording systems digitise for storage and analysis.

Further exploration

Explore

DOSITS — interactive tutorials on sound speed, refraction and the SOFAR channel.

See real observatories

Ocean Networks Canada — cabled, autonomous and mobile observing systems.

Read

Howe et al. (2019), *Observing the Oceans Acoustically*, Frontiers in Marine Science, 6:426.

Explore global monitoring

CTBTO — hydroacoustic monitoring network and station technology.

References

Scientific references and educational resources

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Figure and media credits

- Slide 1: Underwater photograph, Ocean Networks Canada.
- Slide 5: Colladon (1893), historical sound-speed experiment, via DOSITS.
- Slides 6–7: Sound-speed profiles, Webb, Figure 6.4.1, CC BY 4.0.
- Slide 9: SOFAR sound propagation, Webb, Figure 6.4.2, CC BY 4.0.
- Slide 10: Long-range sound propagation infographic, NOAA Fisheries.
- Slide 11: Passive acoustic monitoring technologies, NOAA Fisheries.
- Slides 12 and 15: Hydrophone deployment photographs, CTBTO.
- Slide 14: Hydrophone deployment, U.S. Coast Guard / Tara Molle, via NOAA Ocean Exploration.